# The optimal de-noising algorithm for ECG using stationary wavelet transform

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## Abstract

*The artifacts of ECG signals include baseline wander (BW), muscle (EMG) artifact, electrode motion artifact and power line interference. In order to get the optimal and robust de-noising algorithm among the generally used de-noising methods based on stationary wavelet transform (SWT), we adjust the signal-to-noise ratio (SNR) of the noisy signal from 1db to 10db, and evaluate the results by means of SNR and visual inspection, then conclude using Symlet4, decomposition at level 5, and Hard shrinkage function with empirical Bayesian (EBayes) threshold can get consistently superior de-noising performance. In addition, test the proposed algorithm using MIT–BIH Noise Stress database, the results demonstrate that the proposed method improves the SNR and preserves the waveform, which can be used for clinic analysis.*

## 1. Introduction

In ECG de-noising, the goal is to recover the valid ECG from the undesired artifacts with the minimum distortion for presenting a signal that allows easy visual interpretation and auto diagnosis.

Wavelet transform (WT), due to its sparsity, locality, and multi-resolution natures, has emerged as a simply yet effective de-noising tool. Donoho and Johnstone developed a de-noising methodology called wavelet shrinkage or thresholding [1], much related work has been carried out for applications in de-noising ECG signals. Szi-Wen Chen et al (2006) de-noised ECG signals through wavelet shrinkage (threshold value = 4 × std(detail1), threshold function=hard thresholding) before real-time QRS detection [2].Poornachandra, S. (2008) used an adaptive mean threshold and soft thresholding function to shrink the wavelet coefficients [3]. Due to the selections of shrinkage function and threshold value are crucial to wavelet shrinkage de-noising. So Zhao, Z. et al (2007) compared with three shrinkage functions and five thresholds, finally gave an optimal de-noising scheme based on wavelet shrinkage method: using Non-garrote threshoding function with generalized cross validation (GCV) threshold, and testing this method using ECG signals [4]. Singh, B.N. et al (2006) investigated the optimal wavelet basis function for de-noising ECG signal. They concluded Db8 was the most appropriate wavelet basis function for the ECG de-noising application [5]. Zhang, J. (2005) applied Mallat algorithm to suppress the BW, and the soft-thresholding denoising algorithm to suppress powerline interference and the EMG [6]. Zhang D. (2005) estimated the BW via coarse approximation in DWT, reduced the high-frequency noise via Empirical Bayes (EBayes) posterior median wavelet shrinkage method with level-dependent and position dependent thresholding values [7].

De-noising using Wavelet Shrinkage has a translation variance problem, so the following algorithms tried to solve this problem: Kumari, R.S.S. et al (2008) used Cyclic Shift Tree De-noising technique for reducing white Gaussian noise or random noise, EMG noise and Power line Interference [8]. Su, L. et al (2005) employed Translation-Invariant (TI) method to de-noise ECG signals [9]. Chmelka, L. et al (2005) used combined wavelet shrinkage with wiener filtering, four-level shift-invariant dyadic discrete-time wavelet transform decomposition and hybrid thresholding was used, respectively [10].

According to the literatures in WT de-noising domain, it has been tested that using Cyclic Shift Tree, TI or SWT (based on the same principle) can improve SNR and reduce the Pseudo-Gibbs phenomenon [8-10], which is superior than classical WT. Based on classical WT, Zhao, Z. et al compared with three shrinkage functions and five thresholds, finally gave an optimal de-noising scheme; and Singh, B.N. et al investigated the optimal wavelet basis function for the ECG de-noising application. So, this paper studies the optimal de-noising algorithm for the ECG based on SWT, including the selection of the optimal wavelet

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basis, appropriate thresholding method and decomposition depth, also called level or scale.

## 2. Stationary wavelet transform

The SWT, also be called at TI, Shifted Invariant, is similar to the classical discrete WT (DWT), except the signal is never sub-sampled, and the filters are up-sampled at each level of decomposition. The definition of SWT is to average some slightly different DWT, called $\varepsilon$-decimated DWT [11,12].

Set classical WT decomposition at level $n$ satisfied:

$$\left.\begin{aligned} cA^n &= h * cA^{n-1}, n = 1,2\cdots,N \\ cD^n &= g * cA^{n-1}, n = 1,2\cdots,N \end{aligned}\right\} \quad (1)$$

$N$ is the maximum decomposition level; $h$ and $g$ are low-pass filter and hi-pass filter, respectively; $cA^n$ are the approximate coefficients at level $n$, and $cD^n$ are the detailed coefficients at level $n$, $cA^0$ is the original signal. Then SWT decomposition at level $n$ can be written as the following:

$$\left.\begin{aligned} scA^n_{\varepsilon_1,\cdots\varepsilon_n} &= h^{(n-1)} * scA^{n-1}_{\varepsilon_1,\cdots\varepsilon_{n-1}}, n = 1,2\cdots,N \\ scD^n_{\varepsilon_1,\cdots\varepsilon_n} &= g^{(n-1)} * scA^{n-1}_{\varepsilon_1,\cdots\varepsilon_{n-1}}, n = 1,2\cdots,N \end{aligned}\right\} \quad (2)$$

Actually, in Equation 1., $\varepsilon$ of the classical WT is always 0, shifted by 0, but in Equation 2., $\varepsilon = [\varepsilon_1, \varepsilon_2 \cdots \varepsilon_j]$ , and are adopted 0 and 1 respectively; $h^n$ is low-pass filter, and satisfied: $h^n \uparrow 2 = h^{n+1}, h^0 = h$ , $g^n$ is high-pass filter, and satisfied: $g^n \uparrow 2 = g^{n+1}, g^0 = g$ . So in SWT, $scA^n_\varepsilon$ --the approximate coefficients, and $scD^n_\varepsilon$ -- the detailed coefficients have the same length of the original signal, $scA^0_\varepsilon$ represents the original signal, but the length of the signal must be divided by $2^N$ .

The above scheme is for SWT decomposition, similarly for inverse SWT (ISWT) reconstruction.

## 3. The optimal de-noising algorithm

In general, there are three main steps in WT de-noising procedure [1,12]:

(1) Decompose the noisy signal at level N using a wavelet basis to obtain the approximate and detailed coefficients.

(2) Shrink the coefficients of each level by a thresholding method, including the shrinkage function and the threshold.

(3) Reconstruct the de-noised signal using the shrunk coefficients.

There are many decomposition and reconstruction schemes based on WT, such as wavelet packet, lifting wavelet and SWT. But three points must be addressed whatever using which scheme:

- The appropriate decomposition level.
- The optimal wavelet basis.
- The suitable thresholding method.

The de-noising results are evaluated by means of the SNR and visual inspection. The following is the definition of the SNR:

$$SNR = 20\log\sqrt{sum(f_n{}^2)\big/sum((f_n - \hat{f}_n)^2)} \quad (3)$$

Where $f_n$ is the clean signal, and $\hat{f}$ is the noisy signal or the de-noised signal.

### 3.1. Wavelet basis

According to the key properties of the most commonly used wavelet bases: Morlet wavelet, Mexican hat wavelet, and Meyer wavelet don't have compact support, fast algorithm, and vanishing moments properties; Daubechies wavelets (DbN, N is the order), Symlets (SymN) and Coiflets (CoifN) are orthogonal and compact support wavelets, but DbN are not symmetrical, except Db1, because Db1 is Haar wavelet. SymN and CoifN are near symmetrical. Biorthogonal wavelets (Bior Nr,Nd, r for reconstruction, d for decomposition) have compact support and symmetry properties [12].

Referring to the related literatures in WT de-noising, Db4, Db8, Sym4, Sym7, Sym8, Sym10, Coif5, discrete Meyer wavelet (DMey) and Bior4.4, are suitable wavelet bases for de-noising ECG signals.

Based on the optimal wavelet basis selection, synthetically consider the key properties, and the shape of the scaling function of wavelet bases [12,13]. Symlets are better wavelet bases to process ECG signals theoretically due to their orthogonal, compact support, near symmetry characteristics, and the scaling function has similarity of ECG signals.

### 3.2. Shrinkage functions and threshold selection rules

There are four shrinkage functions commonly used: Soft, Hard, Firm and Non-Negative Garrote.

Soft threshold and Hard shrinkage functions are proposed by Donoho and Johnstone, the equations can be found in Reference 1.. The non-negative garrote shrinkage function was first introduced by Breiman (1995), and the firm shrinkage rule was developed by

Gao and Bruce (1997), the equations can be found in Reference 14. and 15., respectively.

Most commonly used thresholds include: Universal, Minimax, SURE (Stein's Unbiased Risk Estimate), Hybrid, empirical Bayesian (EBayes), generalized cross validation (GCV).

Minimax threshold selection uses minimax principle to choose a fixed threshold to yield minimax performance for mean square error against an ideal procedure. The minimax criterion gives a table of the threshold values for given data sizes that is based on calculations of the minimax risk bound for the wavelet estimate (the table can be found in Reference 16.).

Universal threshold proposed by Donoho as an alternative to the use of minimax threshold, which is bigger in magnitude than the Minimax threshold.

SURE is an unbiased estimator of the mean-squared error of a given estimator. This technique is named after its discoverer, Charles Stein, and it has been used by Donoho and Johnstone to determine the optimal shrinkage factor in a wavelet denoising setting. The threshold is obtained by minimize the unbiased estimate risk.

The above threshold rules can be found in Reference 1. and 16.

Hybrid (heuristic) is a mixture of the two previous options. As a result, if the SNR ratio is very small, the SURE estimate is very noisy. So if such a situation is detected, the fixed form threshold is used [12, 15].

$$\left.\begin{aligned} &\lambda^h = \begin{cases} \lambda^u, & s < \gamma \\ \min(\lambda^s, \lambda^u), & s \geq \gamma \end{cases} \\ &s = \sum_i (x_i^2 - 1) \Big/ n, \gamma = (\log(n)/\log(2))^{3/2} \Big/ \sqrt{n} \end{aligned}\right\} \quad (4)$$

The EBayes method is very good at tracking the minimum of the average square error. The EBayes threshold is always close to the optimal thresholds, and right across the range of sparsity considered—the average square error obtained by the EBayes threshold is very close indeed to the best attainable average square error [17].

A Bayesian model is used for the parameter $f_i$ (de-noised signal). Under this model, each $f_i$ is zero with probability $(1-w)$, while, with probability $w$, $f_i$ is drawn from a symmetric heavy-tailed density $\rho$. The mixing weight $w$ is the key parameter in the prior. It is chosen automatically from the data, using a marginal maximum likelihood approach; Estimation within the Bayesian model is a thresholding procedure, and the choice of $w$ is equivalent to a choice of the threshold $\lambda$.

$$f_{prior}(\theta) = (1-\mathrm{w})\delta_0(\theta) + \mathrm{w}\ \gamma(\theta) \quad (5)$$

The nonzero part of the prior, $\gamma$, is assumed to be a fixed unimodal symmetric density. We concentrate on two particular possibilities for the function, one is Laplace prior and the other is quasi-Cauchy prior.

GCV is another approximation of the MSE, and the GCV threshold $\lambda^g$ is obtained by minimize the GCV estimation [4,18]:

$$\lambda^g = \arg\min GCV(\lambda; x) \quad (6)$$

Above all mentioned thresholds are obtained with the standard deviation $\sigma$ =1, in practice, we only shrink the detail coefficients. Take the median absolute deviation of the wavelet coefficients at the finest scale of resolution and divide by 0.6745 to obtain a crude estimate $\hat{\sigma}$.

## 3.3. Selecting of the optimal de-nosing algorithm

Using 4 shrinkage functions match 6 thresholds, respectively, to compose 24 different thresholding methods (see 3.3). The following are the main steps to realize how to choose the best de-noising algorithm:

(1) Decompose a simulated noisy ECG signal (adding white noise, and the SNR is from 1db to 10db) using SWT with 9 different wavelet bases from level 3 to 8, respectively.

(2) Shrink the coefficients of each level by 24 thresholding methods, respectively.

(3) Reconstruct the de-noised signal, then calculate the SNR of each de-noised signal.

Obtain the optimal de-noising algorithm, including the best decomposition level, the wavelet basis, and the thresholding method, by comparison of the SNR and visual effect.

Simulate a noisy ECG signal with SNR from 1db to 10db, and decompose the signal using SWT at level from 3 to 8, respectively. Table 1. Lists the highest SNR and its corresponding de-noising method. Eg. 5.4D4,h,M represents using Db4 wavelet and Minimax threshold with Hard shrinkage function can get the highest SNR (5.4) at level 3 when original SNR=1db, among the 24 de-noising methods. The following are the abbreviations of wavelet bases, shrinkage function, and threshold Db4(D4), Db8(D8), Sym4(S4), Sym7(S7), Sym8(S8), Sym10(S10), Coif5(C5), Bior4.4(B4), Dmey(M); Hard(h), Softe(s), Firm(f), G(garrot); Universal(U), Minimax(M), SURE(S), Hybrid(H), Bayesian(B), GCV(G).

Table 1. The de-noising algorithm corresponding to the highest SNR at the level 3 from level 8

| L | 1db | 2db | 3db | 4db | 5db | 6db | 7db | 8db | 9db | 10db |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 5.4 D4, h,M | 6.9 D4, h,M | 8.8 B4, h,M | 10.8 B4, H,M | 12.0 B4, h,M | 13.1 B4, h,M | 14.2 B4, h,M | 15.2 S4, h,M | 16.4 S4, h,M | 17.4 S4, h,M |
| 4 | 9.0 B4, h,M | 10.3 B4, h,M | 11.5 S4, h,B | 13.3 S4, h,B | 14.6 S4, h,B | 15.7 S4, h,B | 16.8 S4, h,B | 17.8 S4, h,B | 18.9 S4, h,B | 19.5 S4, h,B |
| 5 | 11.6 S4, h,B | 13.4 S4, h,B | 14.1 S4, h,B | 15.0 S4, h,B | 16.3 D4, h,B | 17.3 D4, h,B | 18.2 S4, h,B | 18.7 S4, h,B | 19.6 S4, h,B | 20.0 S4, h,B |
| 6 | 11.2 S4, g,M | 13.0 S4, h,B | 13.6 S4, h,B | 13.2 S4, h,B | 15.7 S4, h,B | 16.8 S4, h,B | 17.8 S4, h,B | 18.3 S4, h,B | 16.1 S4, h,B | 19.4 S4, h,B |
| 7 | 11.3 S4, g,M | 11.7 S4, h,B | 12.6 S4, h,B | 14.0 S4, g,M | 15.0 S4, h,B | 16.2 S4, h,B | 17.4 S4, h,B | 18.0 S4, h,B | 18.8 S4, h,B | 19.2 S4, h,B |
| 8 | 11.0 S4, g,M | 12.1 S4, h,B | 12.7 S4, g,M | 13.7 S4, g,M | 14.6 S4, g,M | 15.6 S4, g,M | 17.0 S4, h,B | 17.9 S4, h,B | 18.7 S4, h,B | 19.2 S4, h,B |

Through plenty of experiments, at the beginning of decomposition, the SNR increasing followed the level increasing, and at level 5, the SNR is up to the highest, than decreasing followed the level increasing. So we conclude that using SWT decompose the noisy ECG signal at level 5, and employing Sym4 wavelet basis, and Hard shrinkage function with EBayes threshold can get the highest SNR. Figure 1. shows the de-noised signal corresponding to the highest SNR at level 3 to 8 when noisy signal's SNR=1. We can see that decomposition at level 5 also can get the best visual inspection.

### 3.4. Testing the proposed algorithm

Adding BW, EM, and MA artifacts (from MIT-BIH Noise Stress Test Database) and 5db white noise on a clean signal, (Record 100 from MIT-BIH Arrhythmia Database) served as the noisy signal.

Using SWT decompose the noisy signal at level 5, and employing Sym4 wavelet basis, and Hard shrinkage function with Bayesian threshold, in order to get better de-noising result, through many experiments, we found that reduce the approximation of level 8 from the reconstructed signal directly, can improve the SNR and the visual efficiency. Due to using MIT-BIH database whose sampling frequency is 360Hz, so under the sampling theorem, the frequency band of the level 8 is from 0~0.703125Hz, generally, BW's frequency is less than 1Hz. Table 2. lists the SNR comparison, due to the composite artifacts, noisy signal's SNR is negative. But after de-noising, the SNR is improved to

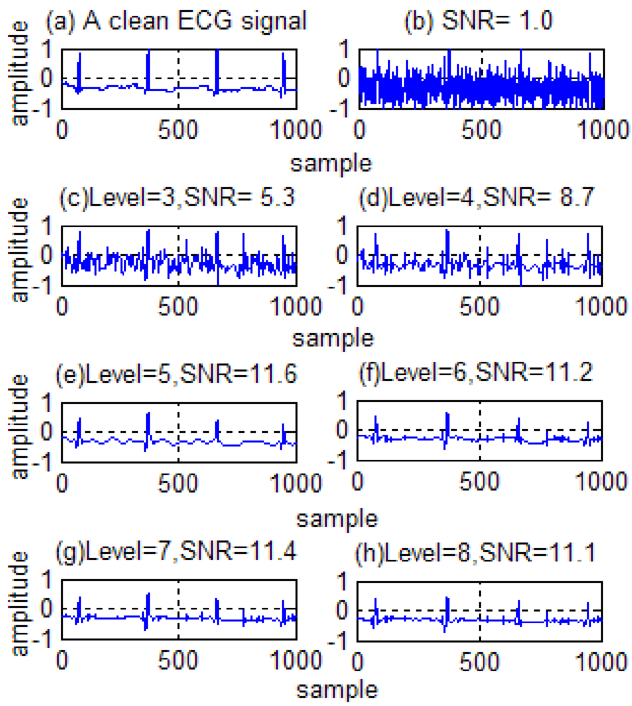


Figure 1. De-noised results from decomposition level from 3 to 8 when noisy signal's SNR=1

positive. Figure 2. makes the visual comparison, we can see noisy signals, Figure 2.(a), Figure 2.(c), and Figure 2.(e) have severer BW( see amplitude), moreover, the valid signal covered by composite artifacts, hardly to diagnose or interpret. But, after de-noising using the proposed algorithm, we can see de-noised signals, Figure 2.(b), Figure 2.(d), and Figure 2.(f) not only reduce the BW, but recover the valid ECG with the minimum distortion.

Table 2. The SNR Comparison

| Signal | SNR (db) | | |
|---|---|---|---|
| Noisy | -13.1 | -19.0 | -22.6 |
| De-noised | 1.1 | 0.9 | 0.6 |

The proposed algorithm is effective in noise removal through using many noisy ECG signals to test. Figure 3.(a) is a real example that cuts 1024 points from the 104 record of MIT-BIH arrhythmias database, which starts in 5 minutes 13 seconds (contains multi-noises). Figure 3.(b) is the de-noised signal using our proposed algorithm. We can see most of the noise is reduced and the valid ECG is recovered.

## 4. Discussion and Conclusion

If the ECG signal only has white noise, we can obtain the same optimal and stable de-noising wavelet basis though the signal's SNR is changing, however, if the signal has composite noise, when SNR is changing,

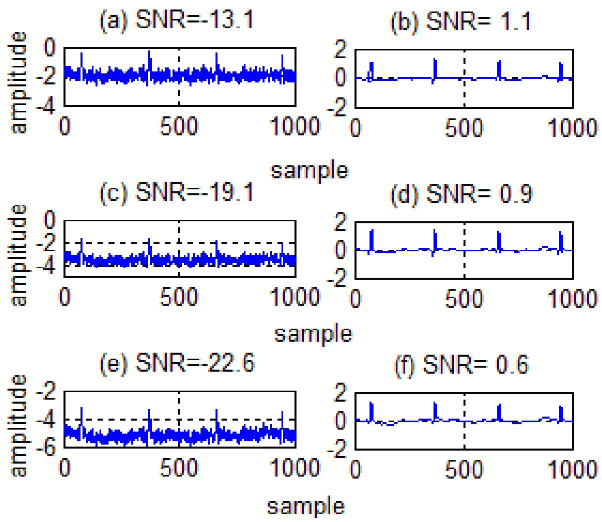


Figure 2. Different noisy signals corresponding to different de-noised signal

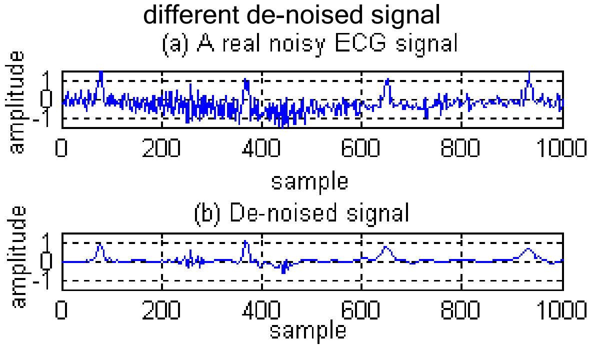


Figure 3. De-noising result of a real example

we can not get the same optimal wavelet basis, which depends on the noise's degree, but there still exist stable results, Sym4 and Db4, but due to Sym4's scaling function has more similarity of ECG signals and it's near symmetry, so still selecting Sym4; as to the thresholding method, Hard shrinkage function can preserve the signal's amplitude though hard procedure creates discontinuity, and EBayes threshold can get better de-noising result throughout the whole de-noising procedure.

About the BW removal, the better way is to reduce the lowest coarse approximation of the wavelet decomposition when reconstruct the de-noised signal.

Through using plenty of MIT-BIH noisy ECG signals to test and simulate, the proposed algorithm is robust and more effective in noise removal, moreover, it can recover the original signal with little distortion.

So, in ECG SWT de-noising domain, using Sym4 decompose at level 5, and Hard shrinkage function with EBayes threshold can get the better de-noising effect before automatic evaluation or clinic analysis.

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